



Reading: Exemplar: Qwiklab: Utilizing Active Directory

Introduction

Active Directory is a core tool for system administrators who need to manage Windows machines. Active Directory allows you to manage users, groups, machines, and the policies that apply to all of them in a centralized fashion.

In this lab, you interacted with Active Directory, used it to add users and groups, edited users memberships, and created a new group policy object (GPO).

This exemplar is a walkthrough of the previous Qwiklab activity, including detailed instructions and solutions. You may use this exemplar if you were unable to complete the lab and/or you need extra guidance in competing lab tasks. You may also refer to this exemplar to prepare for the graded quiz in this module.

Start the lab

You'll need to start the lab before you can access the materials. To do this, click the green **Start Lab** button at the top of the screen.

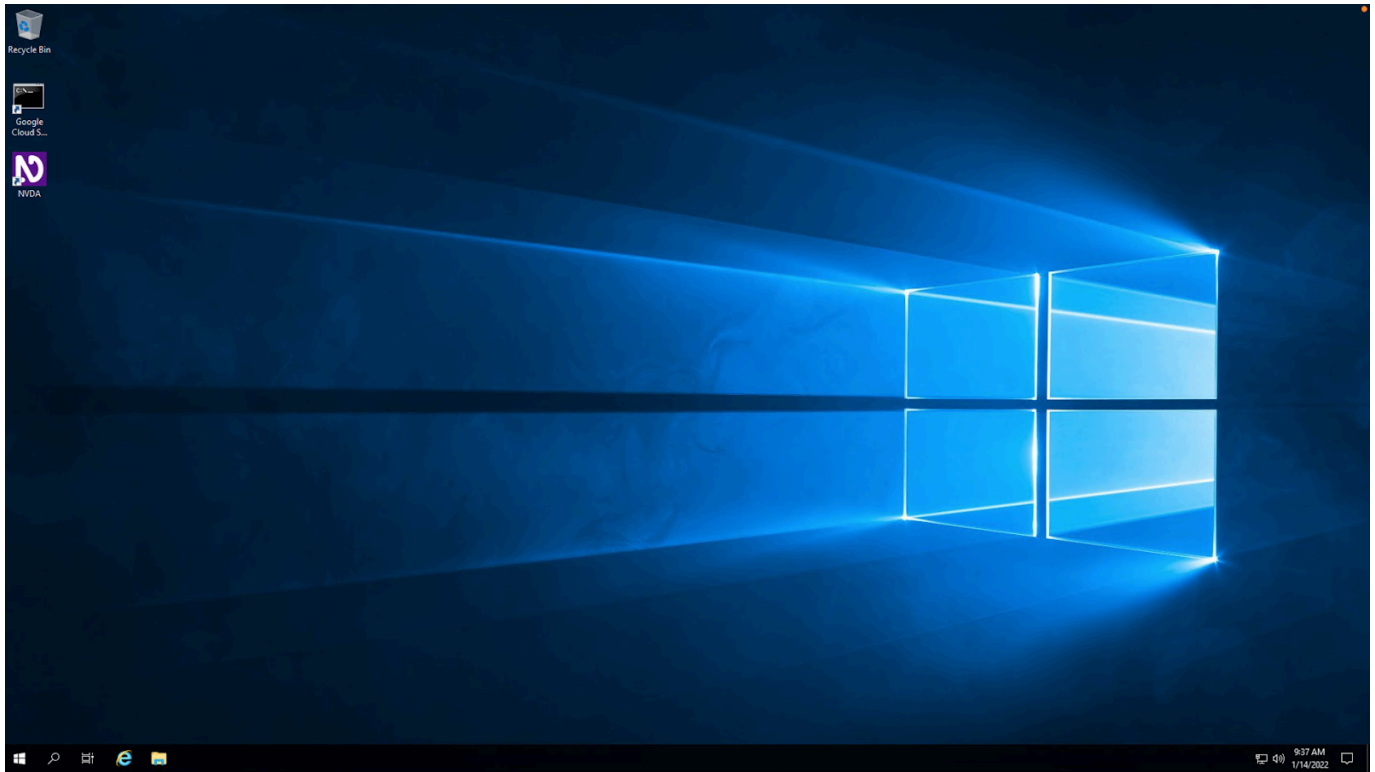
A green rectangular button with rounded corners and the text "Start Lab" in white.

After you click the **Start Lab** button, you will see a panel appear below where the start lab button was that has an **Open Windows VM** button.

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

A light blue rectangular button with rounded corners and a thin blue border, containing the text "Open Windows VM" in blue.

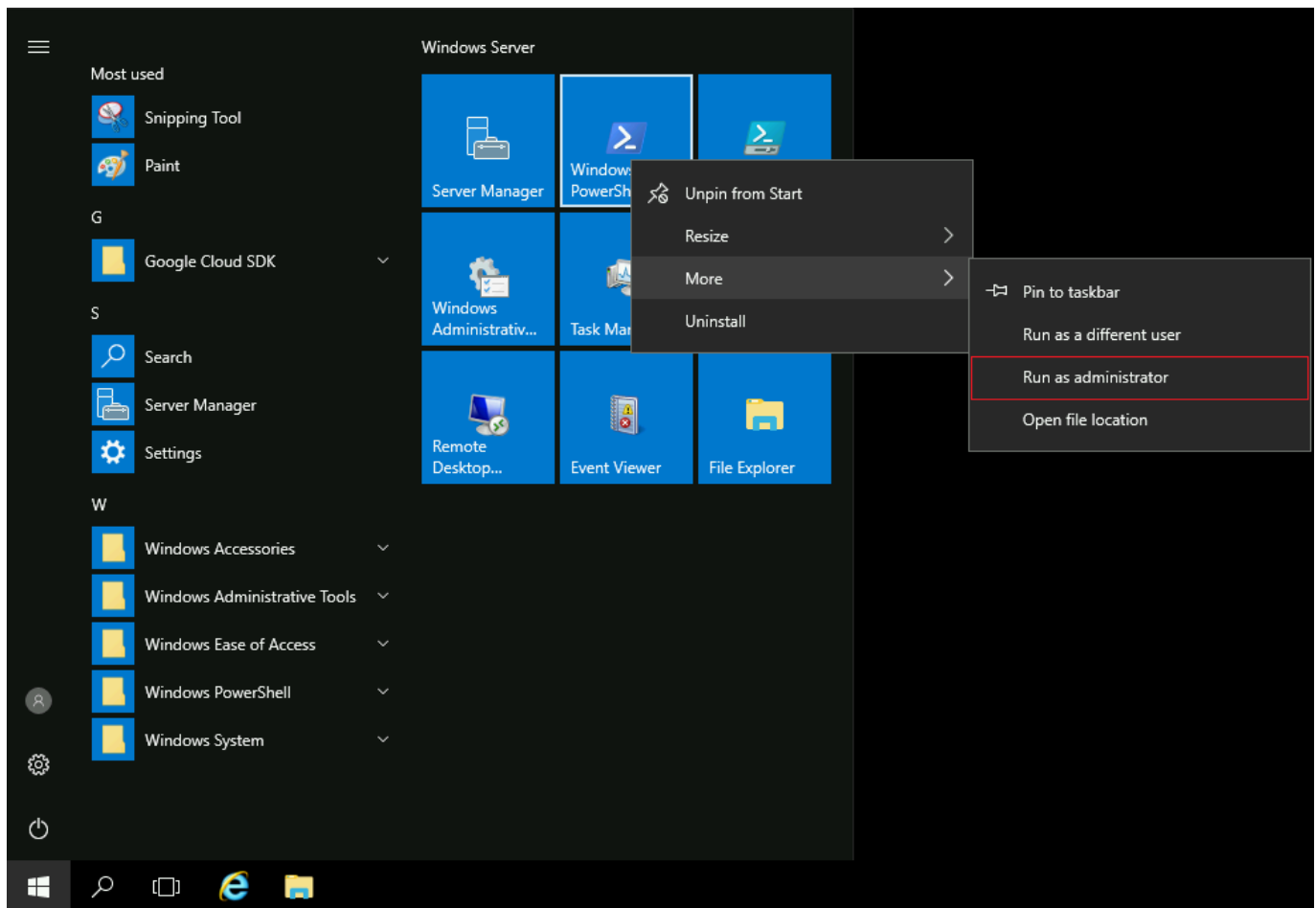
Click the **Open Windows VM** button and a new tab will open with a visual interface for Windows OS, where you will be performing further steps in the lab. You should have a visual interface for Windows that looks like this:



Installing and configuring Active Directory

First, you will need to install and configure **Active Directory**. This is a complicated process, so we have provided PowerShell scripts to automate most of it. Please follow the following instructions carefully.

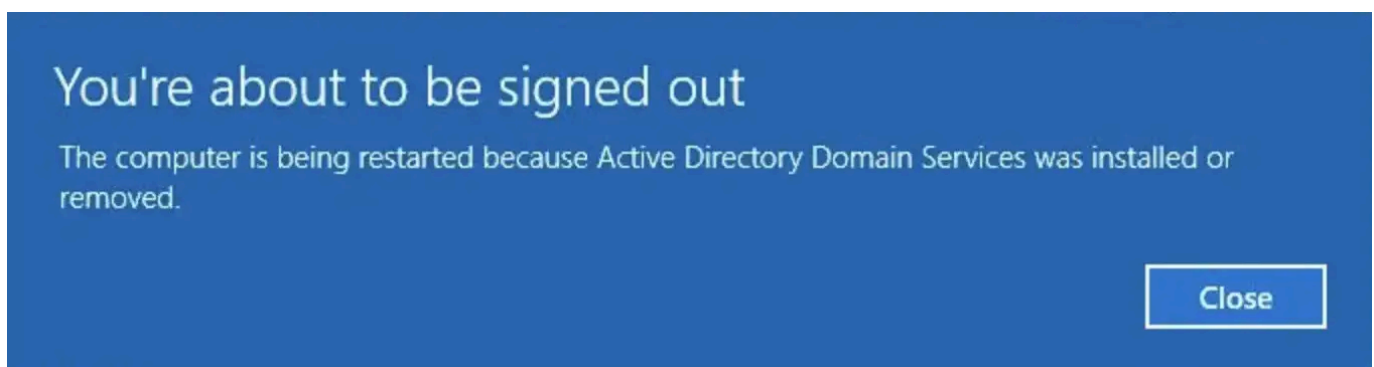
After logging in, open **Windows Powershell** as an Administrator. You can open Windows Powershell as an **Administrator** by opening the **Start Menu**, right-clicking the **Windows Powershell** icon and selecting **More**, then **Run as Administrator**.



In Windows PowerShell, run the following command to install Active Directory:

```
C:\Qwiklabs\ADSetup\active_directory_install.ps1
```

This will run for a couple of minutes. It will print a few warnings, but don't worry, those are expected. When it's done, the script will pop up a message indicating that it will restart the computer.



After successfully installing **Active Directories**, the Windows VM automatically disconnects, and it will display the **RDP connection panel** (shown below):

The image shows a dark blue RDP connection panel. At the top center is a circular icon containing a white monitor and a double arrow. Below this, on the left, are labels for 'Protocol', 'Security', 'Server (:port)', 'Domain (optional)', 'User', and 'Password'. To the right of these labels are corresponding input fields: a dropdown menu for 'Protocol' showing 'RDP', a dropdown menu for 'Security' showing 'AUTO', and five empty text boxes for 'Server (:port)', 'Domain (optional)', 'User', and 'Password'. Below the input fields is a 'Connect!' button. In the bottom left corner, there is a small icon of three people and the text '2.9.2.0', followed by the text 'Hosts management'.

There is no need to enter any information at this point. Simply close this current tab and return to the lab page. Click on **Open Windows VM** again, and it will reconnect you to the same **Windows VM**. This will preserve your previous activity, and you don't need to re-run the Active Directory Installation command again.

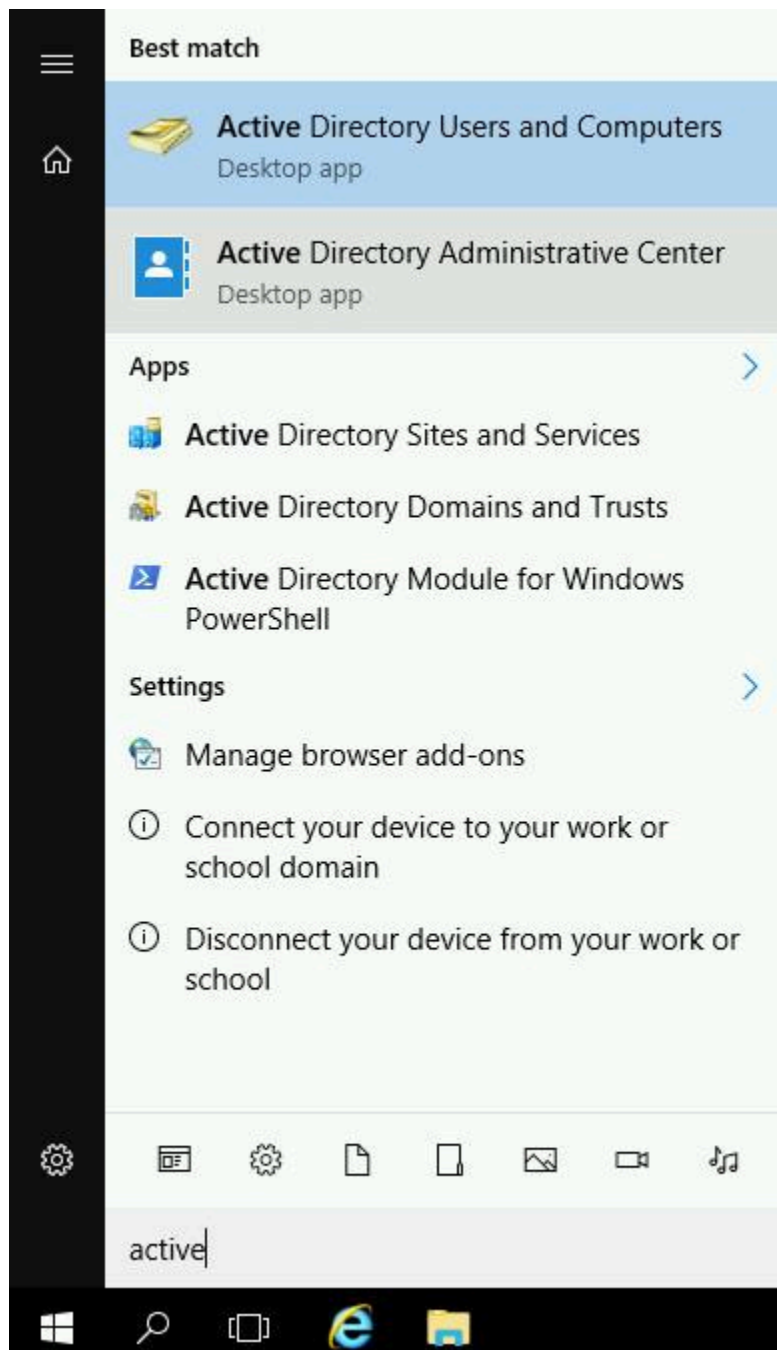
Active Directory has now been installed, but it still needs to be configured. This should be simpler than the previous task. Open **Windows Powershell** as an **Administrator** and run the following command to configure Active Directory and continue with the rest of the lab when it finishes:

```
C:\Qwiklabs\ADSetup\configure_active_directory.ps1
```

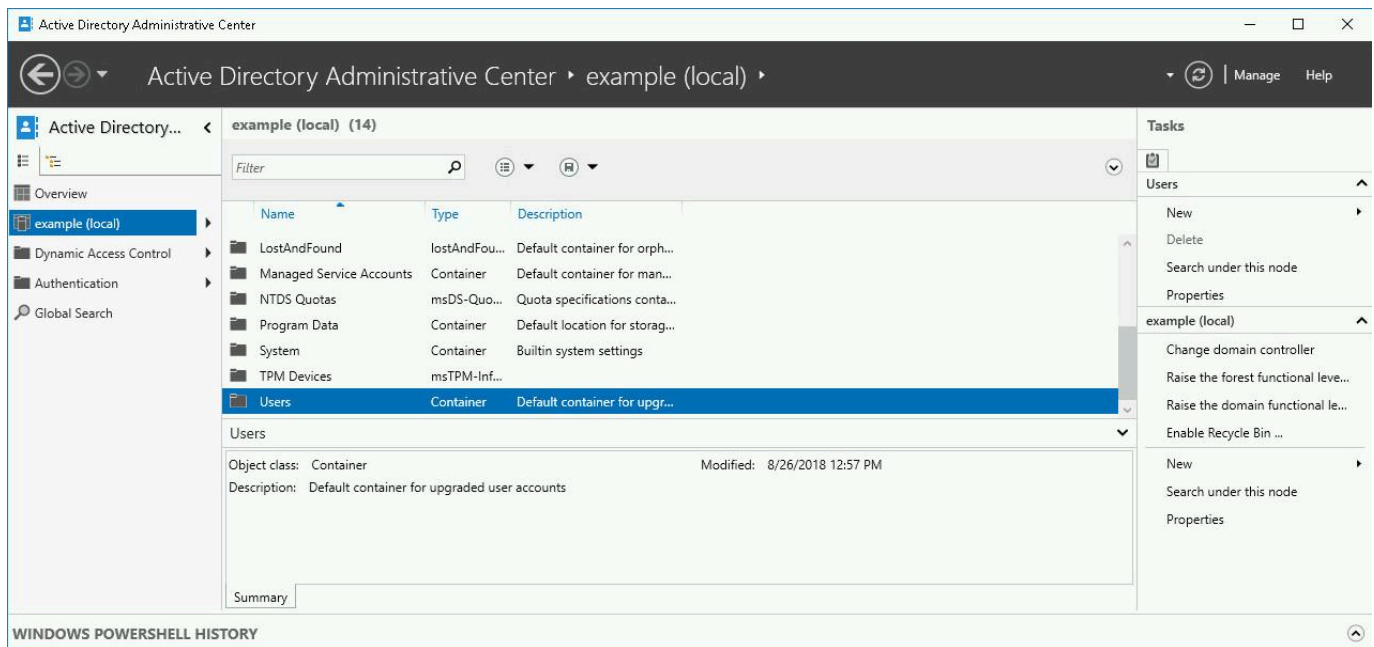
Managing users and groups

Once the above setup is done, you are now ready to experiment with Active Directory.

Open the **Active Directory Administrative Center (ADAC)**. You can find it by typing **active** into the Windows **Start** menu.



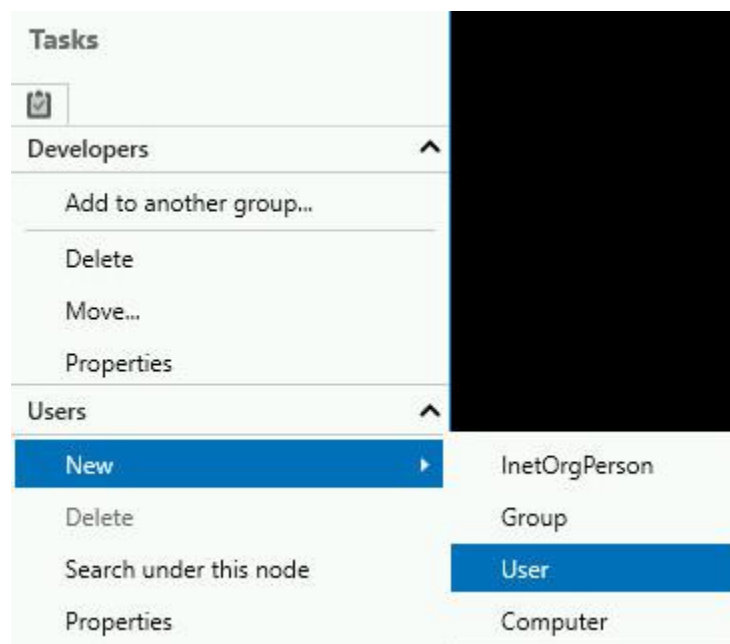
The **Active Directory Administrative Center** allows you to manage your Active Directory installation, by configuring users, groups, computers, and more. Feel free to browse around the resources that already exist in the directory.



For this lab, we want to create a new user called **Alex**. To do that, first click on the **example (local)** entry. This is the entry for the domain that your account can manage. Then scroll down and double click on the **Users** entry to see the list of users and groups that currently exist.

Adding users

To create a new user, take a look at the tasks list on the right. Under the **Users** section, there's a **New** menu entry, which opens a submenu to select what's the type of entity that you want to create. In this case, we want to create a **new user**, so click on the **User** option.



This will open a new window that lets you fill in a number of fields related to the new user. There are a lot of fields available, but only a couple are mandatory (indicated with the red star). You can leave the rest empty. The user that we are creating is called **Alex**, with their username being also **alex**.

Create User: Alex

TASKS ▾ SECTIONS ▾

Account

First name: Alex

Middle initials:

Last name:

Full name: * Alex

User UPN logon: @

User SamAccountName I... example * alex

Password:

Confirm password:

Create in: CN=Users,DC=example,DC=com [Change...](#)

☐ Protect from accidental deletion

[Log on hours...](#) [Log on to...](#)

Account expires: ☒ Never ☐ End of

Password options: ☒ User must change password at next log on ☐ Other password options

☐ Microsoft Passport or smart card is required for interactive log...

☐ Password never expires

☐ User cannot change password

Encryption options: ▾

Other options: ▾

Organization

Display name: Alex

Office:

E-mail:

Web page:

Job title:

Department:

Company:

Manager: [Edit...](#) [Clear](#)

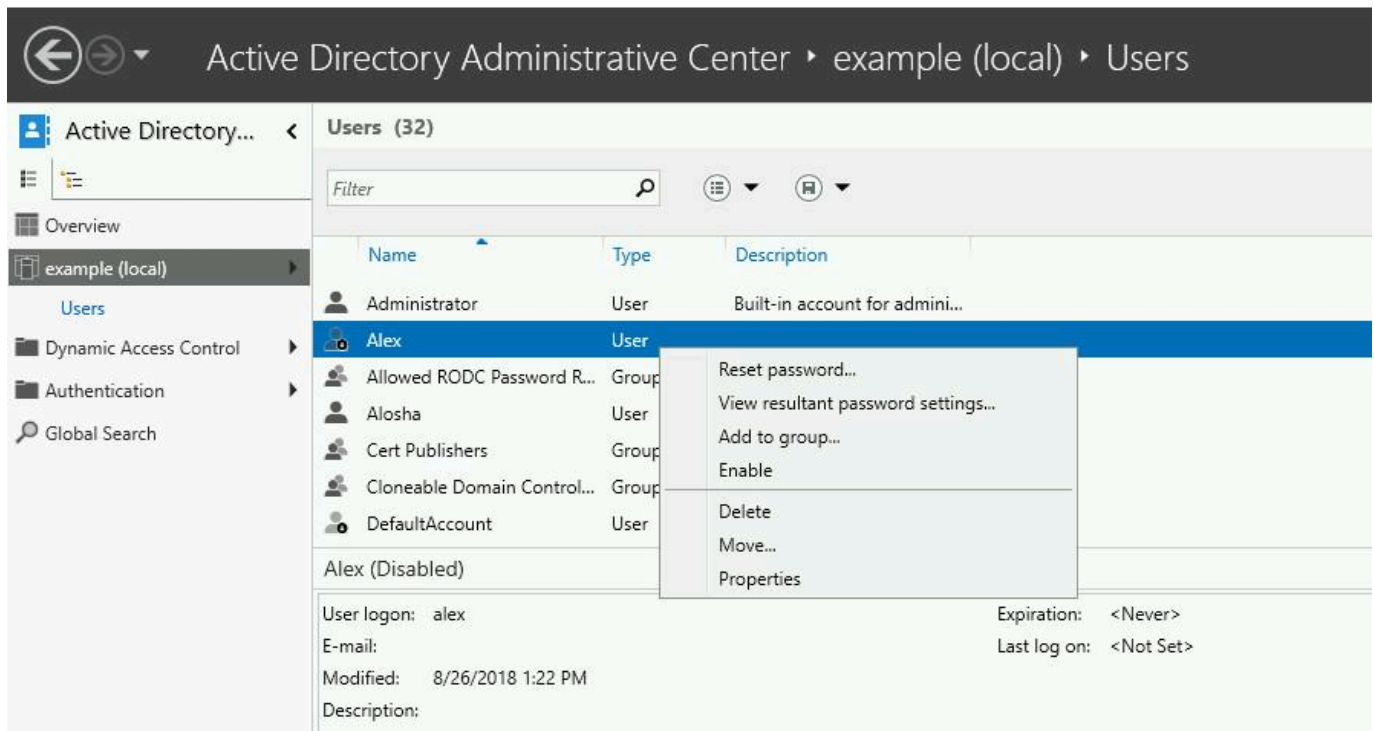
Direct reports:

[Other web pages...](#)

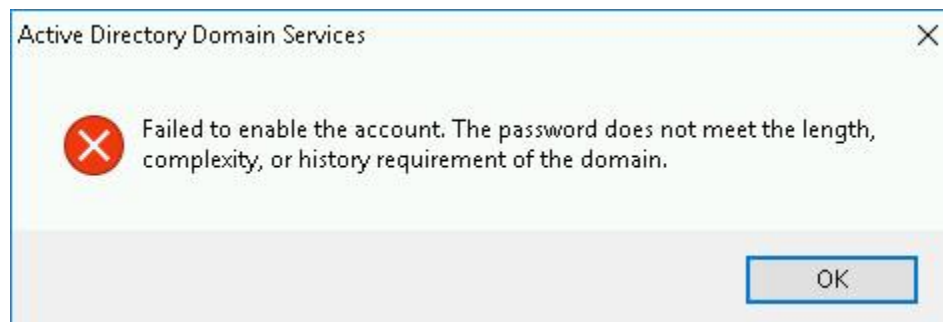
[More Information](#) [OK](#) [Cancel](#)

Once you've entered the necessary data, click the **OK** button to have the user created.

If you click on the newly created account, you will see that where it displays the name of the user, the system says **Alex (Disabled)**.

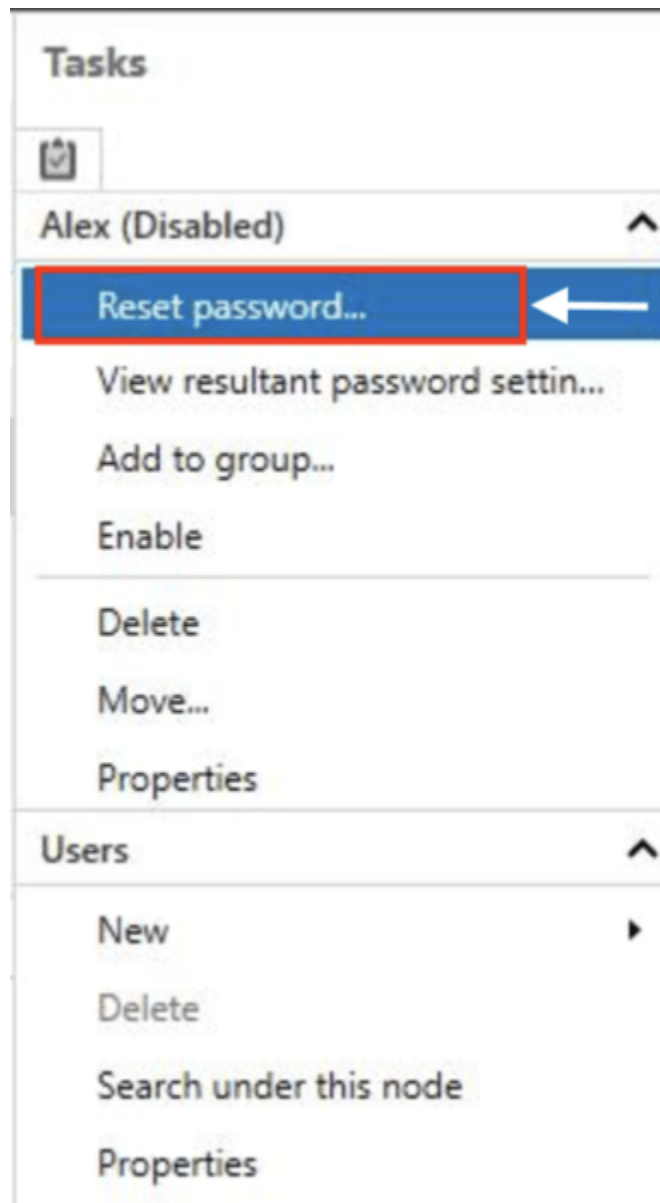


What happens if you right click on the entry and try to **Enable** it?

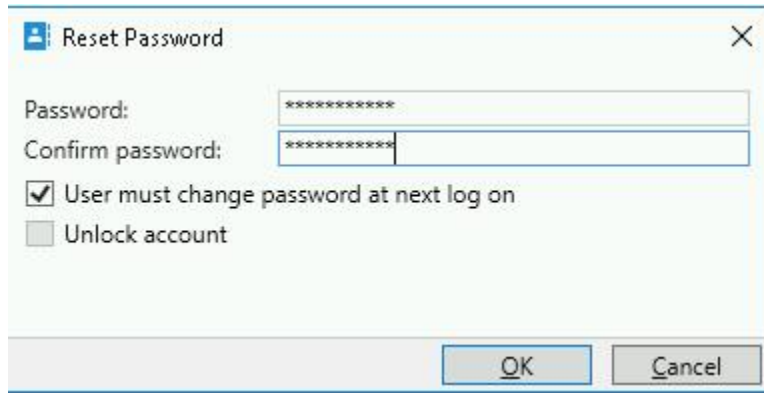


The system will not enable an account that doesn't have a good password. In this case, the password is empty because we haven't set it. Obviously, an empty password is not a good password.

You can set a password using the **Reset password** menu option.



Enter the password and confirm password into the **Reset Password** window. When **User must change password at next login** option is already checked, we ensure that the user will change their password when they log in. Now click on the OK button to set the password. The goal of this is that after they've logged in once, the system administrator will not know their new password.

A Windows-style dialog box titled "Reset Password" with a close button (X) in the top right corner. It contains two text input fields: "Password:" and "Confirm password:", both filled with asterisks. Below the fields are two checkboxes: "User must change password at next log on" (checked) and "Unlock account" (unchecked). At the bottom right are "OK" and "Cancel" buttons.

Reset Password

Password: *****

Confirm password: *****

☒ User must change password at next log on

☐ Unlock account

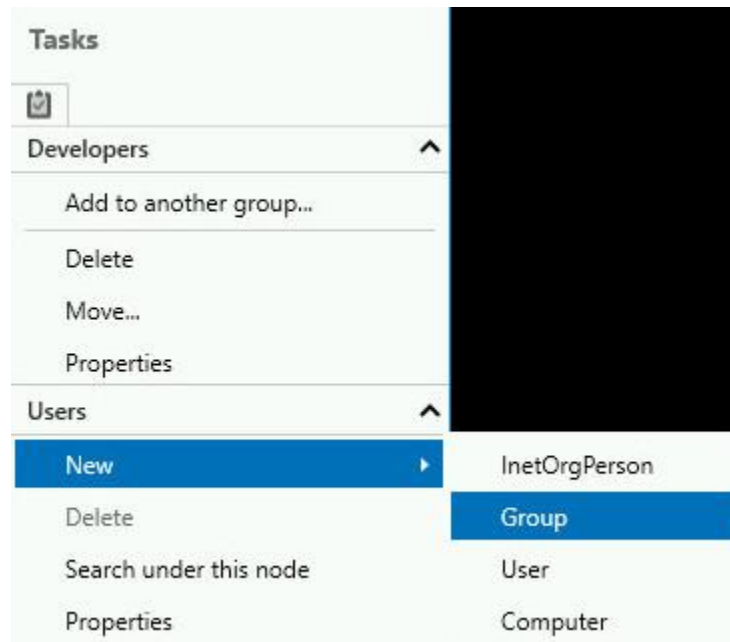
OK Cancel

Once you've set a good password, you can retry enabling the account. This time it should work.

Adding groups

Let's now add a **new group**. If you browse through the existing groups, you will see that there's a group called **Developers** and a group called **Java Developers**. We now want to add an additional group, called **Python Developers**. Add the new group to the Developers group, then add the account we just created for Alex to the Python Developers group.

To create a new group, use the same menu that you used for creating a new user, but this time select the new **Group** option.



This will open a similar window to the one that we saw before, but this time it requires the data for the **Group** rather than the **user**.

Create Group: Python Developers

TASKS ▾SECTIONS ▾

Group

Managed By

Member Of

Members

Password Settings

Group

Group name: * Python Developers

Group (SamAcc... * Python Developers

Group type: ☒ Security ☐ Distribution

Group scope: ☐ Domain local ☒ Global ☐ Universal

☐ Protect from accidental deletion

E-mail:

Create in: CN=Users,DC=example,DC=com

Description: [Change...](#)

Notes:

Managed By

Managed by:

Edit... Clear

☐ Manager can update membership list

Phone numbers:

Main:

Mobile:

Fax:

Office:

Address:

Street

City State/Province Zip/Postal c... Country/Region:

Member Of

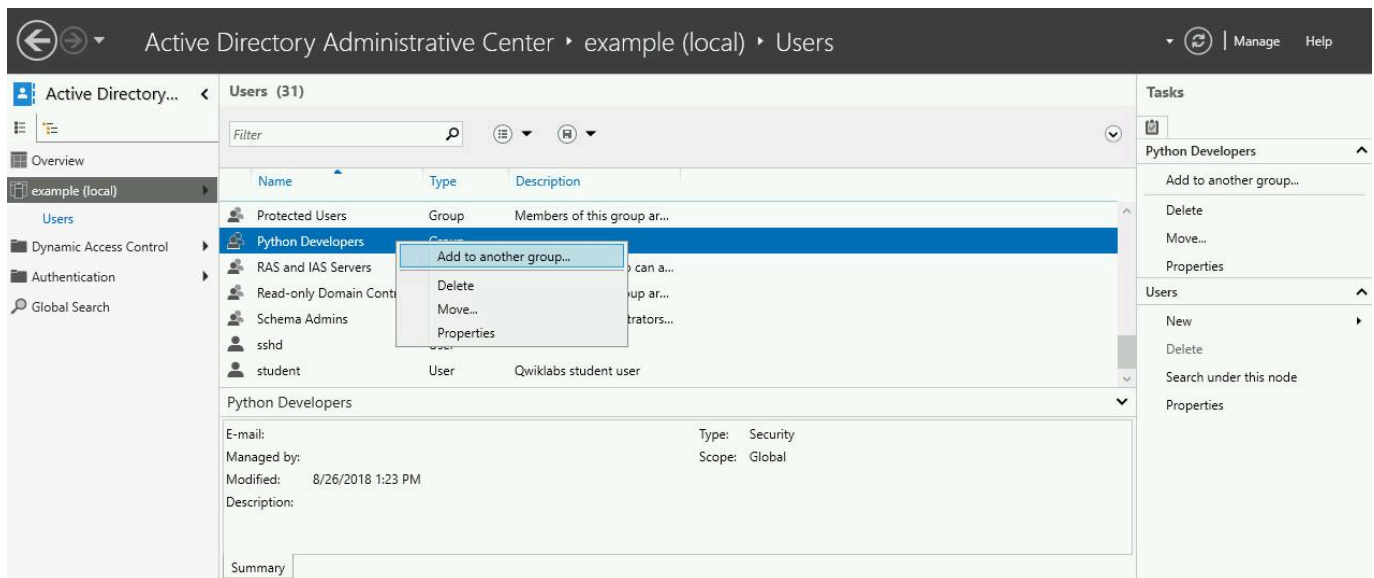
More Information

OK Cancel

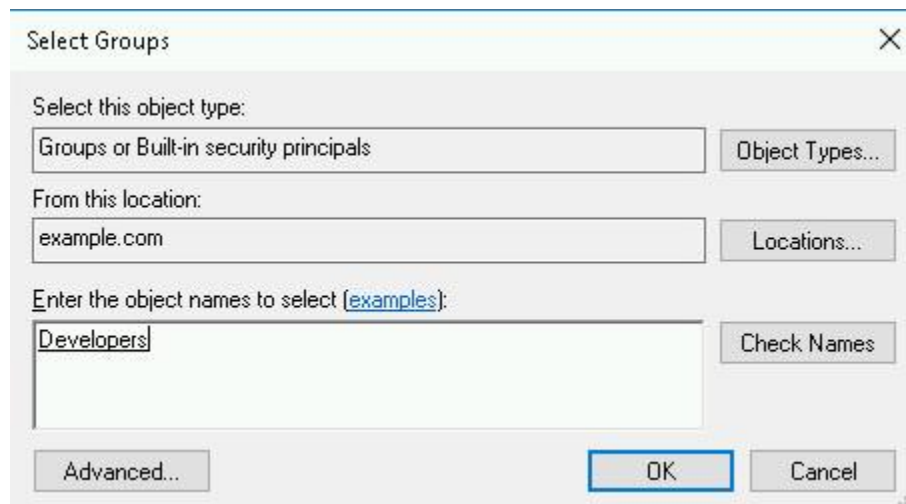
We are creating a group called **Python Developers** and that's the only data that is mandatory. You can also add additional information in the **Description** and **Notes**, if you want. Once you are done, click **OK** to have the group created.

Adding entities to groups

We have a **Python Developers** group, now we want to add it to the **Developers** group that already exists. We can do this by scrolling down to the new entry and then right clicking on the entry in the list and selecting the **Add to another group entry**.



This will open a small window where we need to enter the name of the group. In this case, the group is called **Developers**.



You can use the **Check Names** button to verify that you have entered the name correctly. If you have, it will underline the text. If the name is incorrect, it will show a window saying "Name Not Found."

Clicking the **OK** button will add the **Python Developers** group to the **Developers** group. We now want to do the same for adding **Alex** to **Python Developers**. But we'll follow a different path.

In this case, we will double click the **Python Developers** entry in the list, which will open up an editing window for the group.

Python Developers

TASKS ▾ SECTIONS ▾

Group

Managed By

Member Of

Members

Password Settings

Extensions

Group

Group name: * Python Developers

Group (Sa... * Python Developers

Group type: ☒ Security ☐ Distribution

Group scope: ☐ Domain local ☒ Global ☐ Universal

☐ Protect from accidental deletion

E-mail:

Description:

Notes:

Managed By

Managed by: Edit... Clear

☐ Manager can update membershi...

Phone number

Main:

Mobile:

Fax:

Office:

Address:

City State/Pr... Zip/Post...

Country/Region: ▾

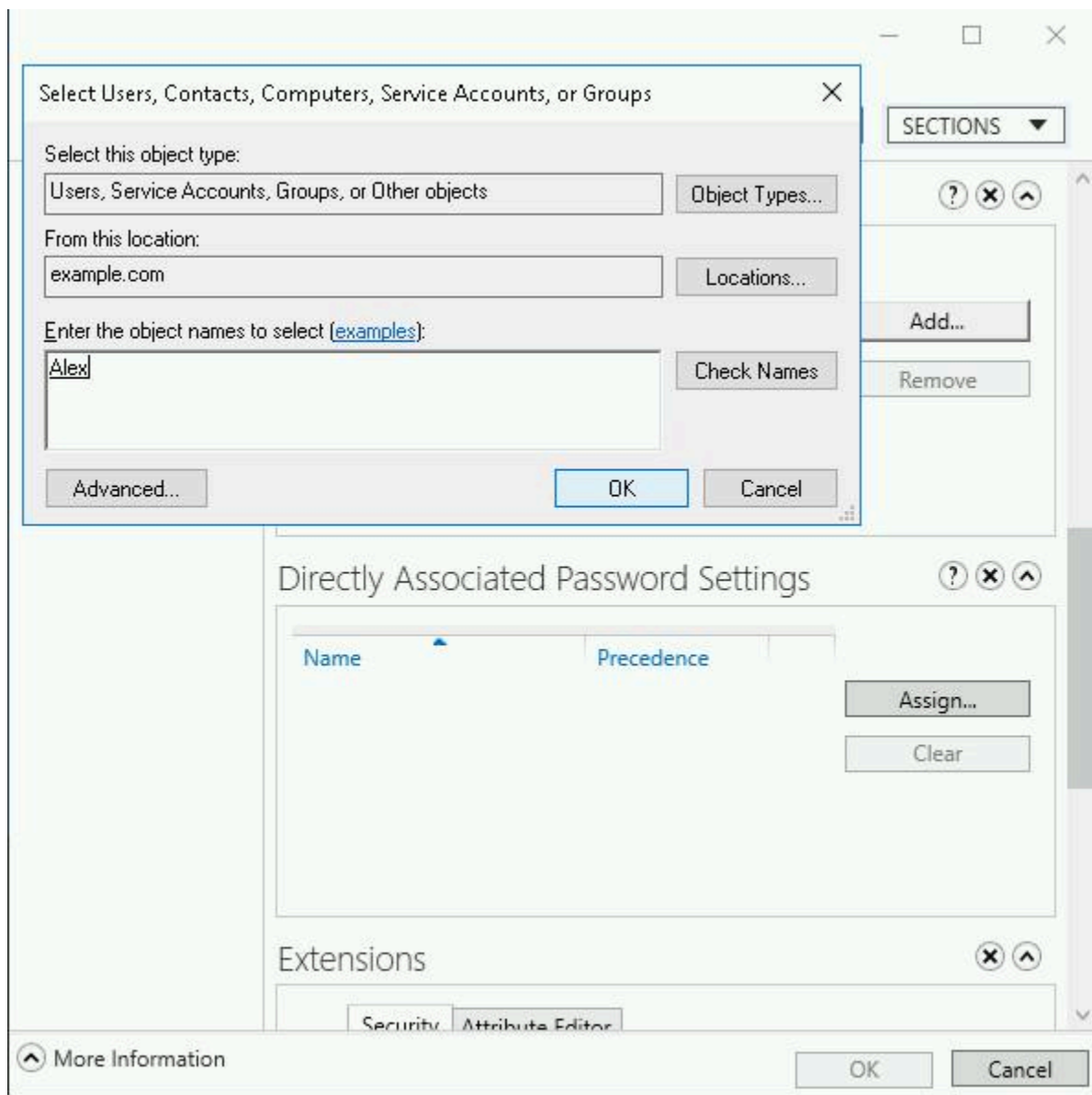
Member Of

Filter 🔍

More Information

OK Cancel

You can scroll down until you find the **Members** section of this window, or you can click on the **Members** link on the left. This section allows us to manually add or remove members from the group.



In this case, what we want to do is to add **Alex** to the group, so click the **Add** button, enter **Alex** in the text field and then **OK** for the addition and **OK** for saving the changes. We've successfully added a new member, Alex, into the group!

Editing memberships

Finally, there's an existing user called **Alosha** that has switched from programming in Java to programming in Python. We want to remove this user from the **Java Developers** group and add them to the **Python Developers** group.

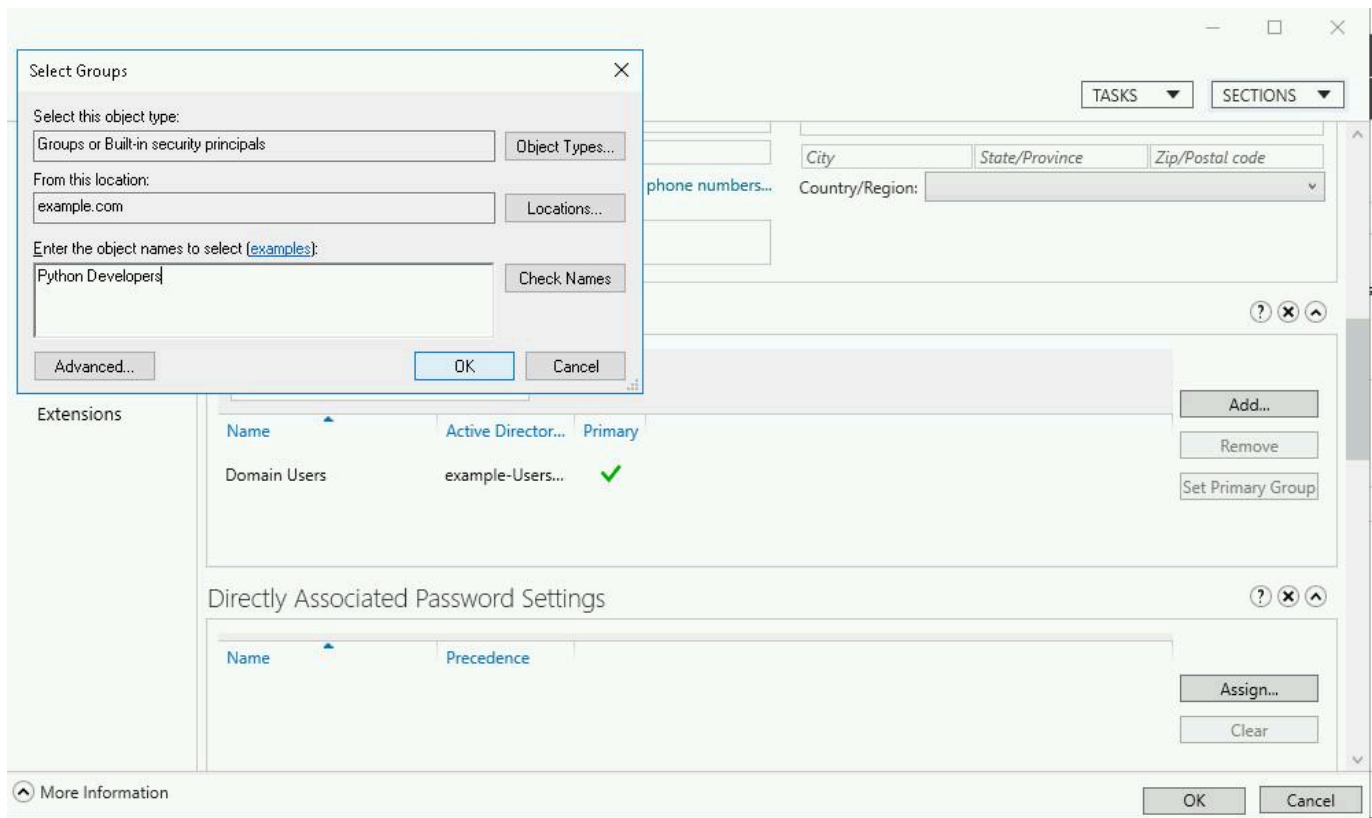
To do this, look for the user **Alosha** in the list and double click on the entry. This will open the properties of the user that you will be able to edit. There's a lot of configuration to each user, click on the section on the left called **Member Of**.

The screenshot shows the 'Alosha' user profile settings window. The left sidebar contains navigation links: Account, Organization, Member Of, Password Settings, Profile, Policy, Silo, and Extensions. The main content area is divided into sections. The 'Member Of' section is active, displaying a table of groups. The table has columns for 'Name', 'Active Director...', and 'Primary'. Two groups are listed: 'Domain Users' (example-Users...) with a green checkmark in the 'Primary' column, and 'Java Developers' (example-Users...). The 'Java Developers' row is highlighted in blue. To the right of the table are buttons: 'Add...', 'Remove', and 'Set Primary Group'. Below the table is a section for 'Directly Associated Password Settings' with an empty table and 'Assign...' and 'Clear' buttons. At the bottom are 'OK' and 'Cancel' buttons.

Name	Active Director...	Primary
Domain Users	example-Users...	✓
Java Developers	example-Users...	

We can see that Alosha is a member of the **Domain Users** group (all users of the domain are members of this group) and of the **Java Developers** group. You can select the **Java Developers** entry and click the **Remove** button to remove that group.

Then click on the **Add** button to add a new membership.

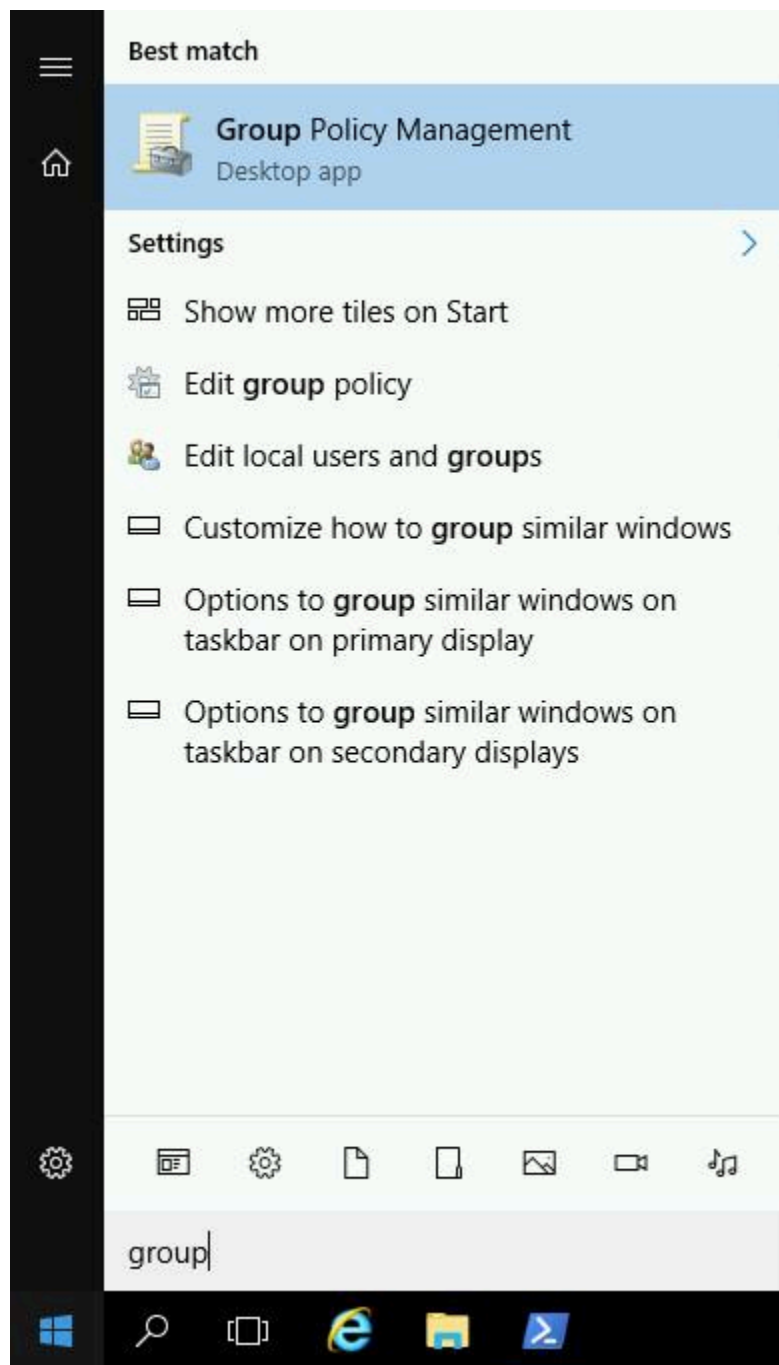


This will pop-up a small window where you need to enter the name of the group that you want to add, in this case **Python Developers**. Once you are done, click **OK** in the **Select Groups** window and then **OK** in the editing user window.

With that, we've created users and groups and we've added and removed group memberships using **Active Directory**. Let's now look into how to manage group policies.

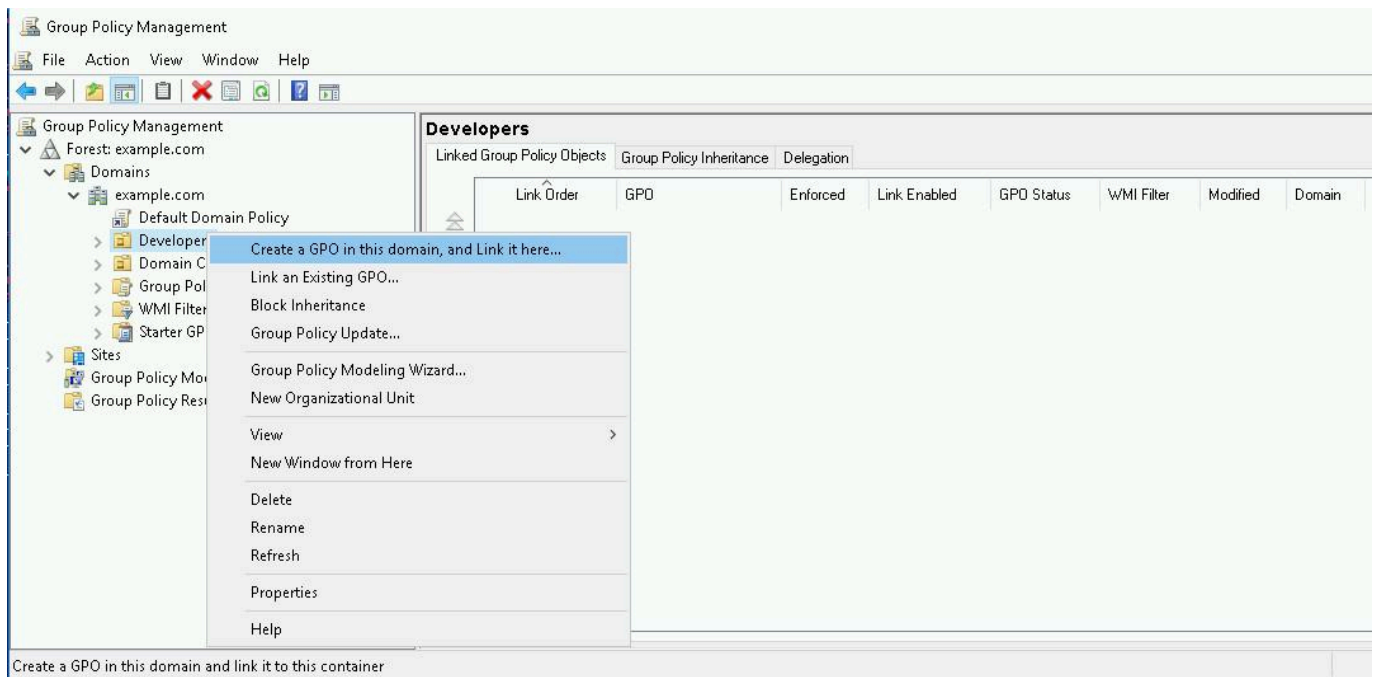
Managing group policies

To manage group policies, we need to use the **Group Policy Management** application. You can find it by typing **group** into the Windows **Start** menu.



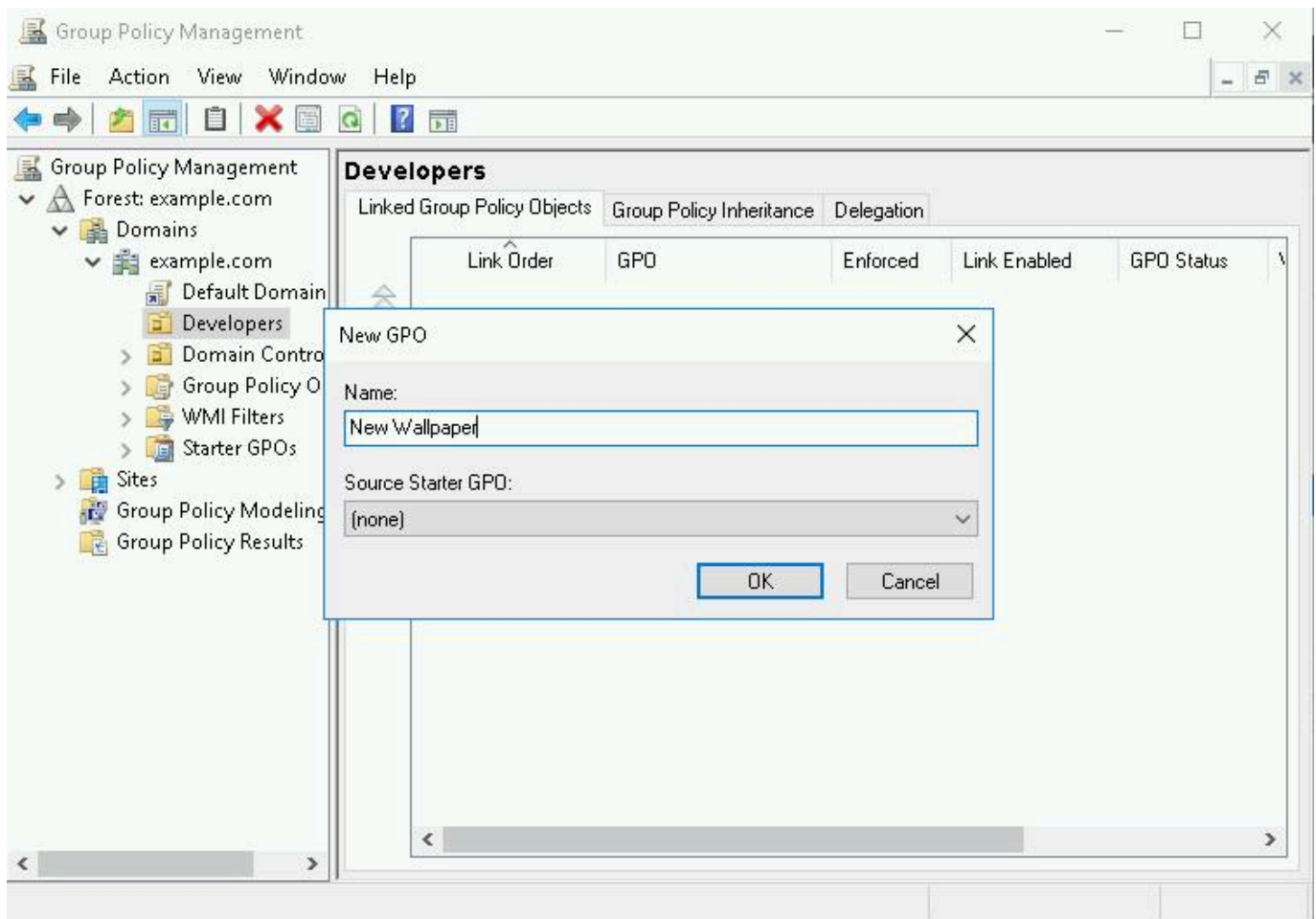
This application allows you to set policies that will manage the way machines in your domain behave. You can apply these policies to the whole domain or to separate **Organizational Units (OUs)**.

In our case, we want to add a new policy to the **Developers** OU that already exists in the domain. To do that, expand the tree until you reach the **example.com** domain tree and find the **Developers** OU inside it.



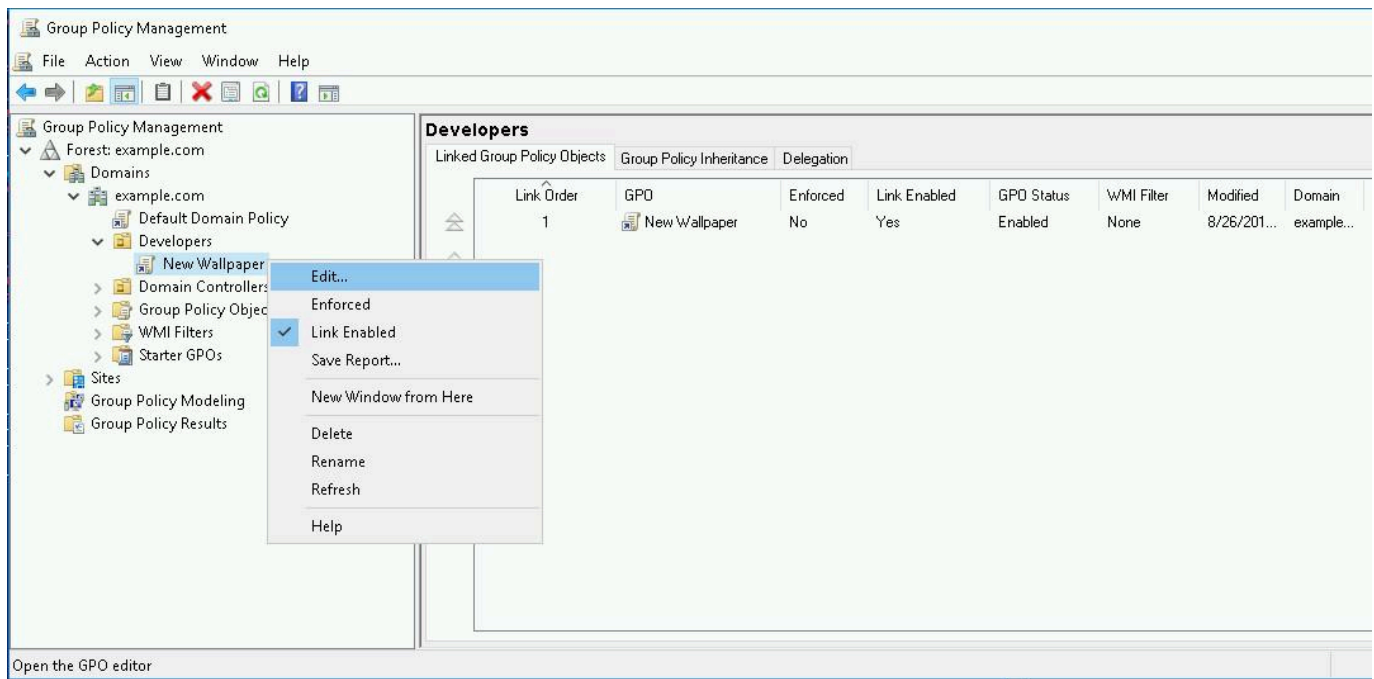
To create a new policy, right click on the **Developer** option and select the first menu entry: **Create a GPO in this domain and Link it here.**

When you click this option, you will be prompted to set a name for the policy and once you do, the policy will get added to the OU.



We want to set a default wallpaper for the machines in the **Developers** OU, so we will call our policy **New Wallpaper**.

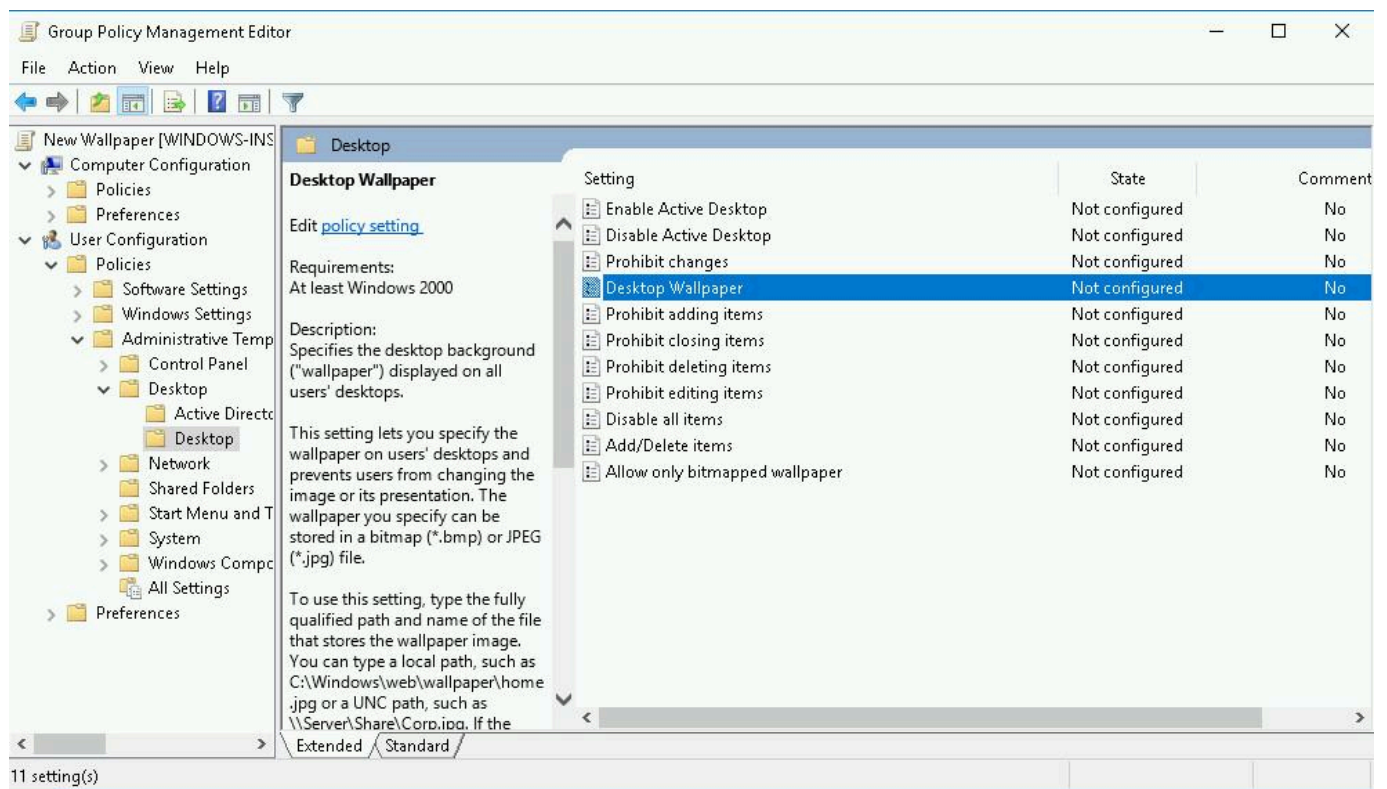
Once created, we want to edit the policy, to do this, right-click on the entry and click on the first menu entry: **Edit**.



Note: You may get a warning message about what linking policies means. That's ok, you can just accept the warning and move on.

This will open a new application: the **Group Policy Management Editor**. This application allows you to navigate and configure all settings that can be set in a group policy.

As we want to set the wallpaper, we need to navigate to this setting by going to: **User Configuration > Policies > Administrative Templates > Desktop > Desktop**.



This opens a list of possible settings that we can configure, including the Desktop Wallpaper. To set the wallpaper to a specific value, double-click on the **Desktop Wallpaper** entry.

Desktop Wallpaper

Previous Setting Next Setting

☐ Not Configured ☒ Enabled ☐ Disabled

Comment:

Supported on: At least Windows 2000

Options:

Wallpaper Name:

C:\qwiklabs\wallpaper.jpg

Example: Using a local path:
C:\windows\web\wallpaper\home.jpg

Example: Using a UNC path:
\\Server\Share\Corp.jpg

Wallpaper Style: Center

Help:

Specifies the desktop background ("wallpaper") displayed on all users' desktops.

This setting lets you specify the wallpaper on users' desktops and prevents users from changing the image or its presentation. The wallpaper you specify can be stored in a bitmap (*.bmp) or JPEG (*.jpg) file.

To use this setting, type the fully qualified path and name of the file that stores the wallpaper image. You can type a local path, such as C:\Windows\web\wallpaper\home.jpg or a UNC path, such as \\Server\Share\Corp.jpg. If the specified file is not available when the user logs on, no wallpaper is displayed. Users cannot specify alternative wallpaper. You can also use this setting to specify that the wallpaper image be centered, tiled, or stretched. Users cannot change this specification.

If you disable this setting or do not configure it, no wallpaper is displayed. However, users can select the wallpaper of their choice.

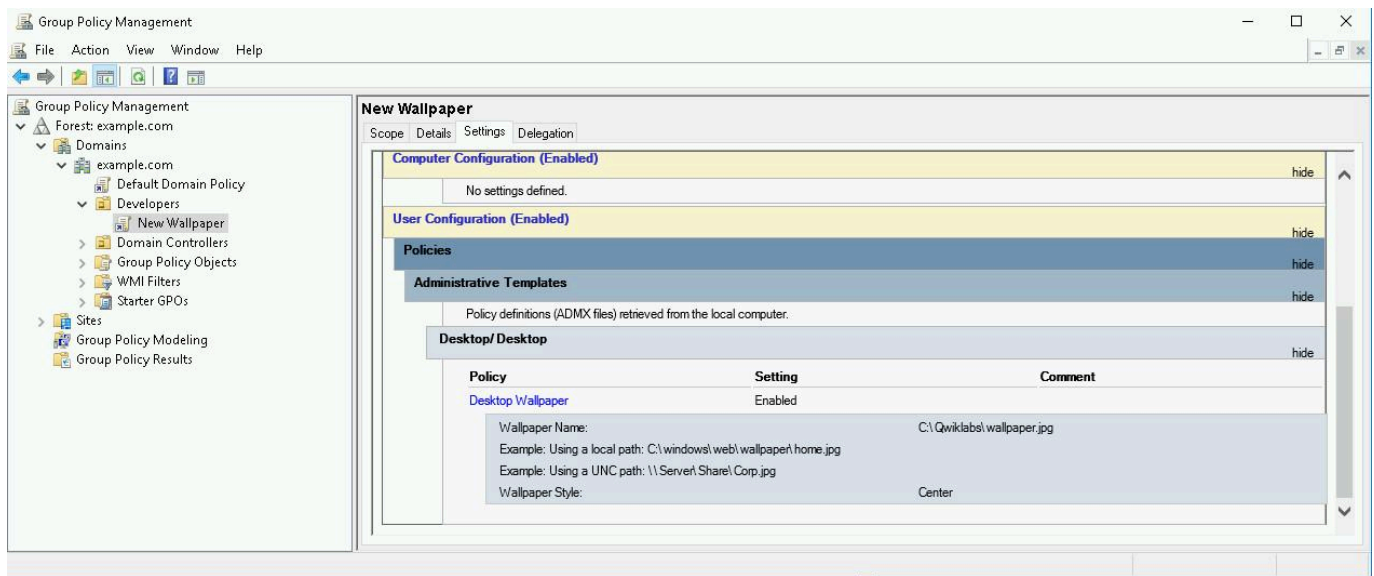
OK Cancel Apply

The window that opens allows you to set the value of the wallpaper. To do that, first click on the **Enabled** button and then enter a path for the wallpaper. The path could be a local path in the machine or a network path on a server that shares files.

For this lab, simply enter **C:\Qwiklabs\wallpaper.jpg** in the section **Wallpaper Name**.

Once you click **OK**, the group policy is created and contains the values we want. To verify this, go back to the **Group Policy Management** application and click the **Settings** tab of the new policy.

Note: This may show a warning that the application needs to be allowed to generate web content. You will need to **Add** the application as a trusted website in order to view its contents.



By clicking the **show** links in the webpage, you can see that the policy has been defined and that the only setting being modified is the Desktop Wallpaper, which is set to the value we set above.

Conclusion

You've now seen how to manage users, groups and group policies using Active Directory. There's a lot more to learn about AD, but these skills are the building blocks for administering a fleet of Windows computers.

Keep it up!
